American Computer Science League

2024-2025 ● Contest 1: Rings ● Intermediate Division

PROBLEM: ACSL Rings is a 3-player game where the object is to score points by tossing a ball through at least one of the 5 rings. Points are awarded as follows:



- Through the aqua or red ring - 1 point
- Through the orange or green ring - 3 points
- Through the black ring - 6 points
- Through the space where 2 rings overlap - the sum of the points for the 2 rings plus 1

We will use a single letter to represent each of the above colors: aqua (A), red (R), orange (O), green (G) and black (B). The overlapping areas will be given by 2 letters in the order shown: aqua and orange (AO), orange and black (OB), black and green (BG), and green and red (GR).

You will be given 3 strings representing each player's results for all tosses that go through at least one ring. The result of each toss will be separated by a single space. Output the 3 players' scores in descending order separated by a single space. If any of the players have the same score, it doesn't matter which one of the tied scores appears first.

EXAMPLE:

Input	`GR A B`
	`OB BG AO O`
	`R G`

Output	`28 12 4`
Explanation	Player 1's tosses go through the green & red, aqua, and black rings for a score of (3+1+1) + 1 + 6 = 12. Player 2 has a score of (3+6+1) + (6+3+1) + (1+3+1) + 3 = 28. Player 3 has a score of 1 + 3 = 4. The output is the string `28 12 4`, the scores in descending order.

INPUT: The input will consist of 3 strings as described above. Each string will be no longer than 100 characters.

OUTPUT: Output the three scores in descending order, each separated by a single space.

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问题：ACSL Rings 是一种 3 **人游戏，游戏**规则**是向** 5 **个圆环投球，球穿过**至少**一个圆环**即可**得分。计分规则如下：**

- **球穿过蓝色或红色圆环——得** 1 **分**
- **球穿过橘色或绿色圆环——得** 3 **分**
- **球穿过黑色圆环——得** 6 **分**
- **球穿过两**个圆**环**的**重叠区域——得分为两**个圆环的得**分之和再加** 1

我们用单个字母来表示上述**每种颜色的圆环：蓝色（**A**）、红色（**R**）、橘色（**O**）、绿色（**G**）、黑色（**B**）。**两个圆环的**重叠区域用两个字母来表示，**字母顺序如下：**蓝色和橘色圆环重叠区域（**AO**）、橘色和黑色**圆**环重叠区域（**OB**）、黑色和绿色圆环重叠区域（**BG**）、绿色和红色圆环重叠区域（**GR**）。**

给定 3 **个字符串分别**表示每位玩家投球穿过至少一个圆环的所有结果。**每次投球穿过**的**圆环**之间**用一个空格隔开。将** 3 **位玩家的得分**按照**降序输出为一个字符串，且每位玩家的得分之间用一个空格隔开。如果任意玩家**之间出现相同**得分，**则相同得分在输出时不论先后顺序。

示例：

输入	`GR A B`
	`OB BG AO O`
	`R G`

输出	`28 12 4`
解析	1 号**玩家投球穿过绿色和红色**圆**环**的**重叠区域、蓝色圆环、黑色圆环，得分为** (3+1+1) + 1 + 6 = 12。 2 号**玩家的得分为** (3+6+1) + (6+3+1) + (1+3+1) + 3 = 28。 3 号**玩家的得分为** 1 + 3 = 4。 将所有**玩家的得分**按照降序**输出为一个字符串：**`28 12 4`。

输入：输入**如上所述的** 3 **个字符串。每个字符串不超过** 100 **个字符。**

输出：按照降**序输出** 3 **位玩家的得分，且每位玩家的得分之间用一个空格隔开。**

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SAMPLE INPUT:

```
1. GR A B
   OB BG AO O
   R G
2. A R O G B
   A A B B
   R O B
3. AO OB B G A B BG GR
   R R G G AO AO BG BG B
   AO GR A B R G O BG OB
4. A A A B B R O O O O
   BG BG GR GR O O O A B
   BG BG GR GR O O O B B G
5. A B B O O G AO OB GR BG
   GR GR BG BG OB AO AO
   B B B G G G A A A GR AO
6. GR BG OB AO A G B
   A O B AO GR GR BG O GR AO
   O O B G A A GR OB AO BG
```

SAMPLE OUTPUT:

```
1. 28 12 4
2. 14 14 10
3. 46 44 44
4. 54 46 28
5. 52 50 40
6. 48 47 40
```

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样例输入：

```
1. GR A B
   OB BG AO O
   R G
2. A R O G B
   A A B B
   R O B
3. AO OB B G A B BG GR
   R R G G AO AO BG BG B
   AO GR A B R G O BG OB
4. A A A B B R O O O O
   BG BG GR GR O O O A B
   BG BG GR GR O O O B B G
5. A B B O O G AO OB GR BG
   GR GR BG BG OB AO AO
   B B B G G G A A A GR AO
6. GR BG OB AO A G B
   A O B AO GR GR BG O GR AO
   O O B G A A GR OB AO BG
```

样例输出：

```
1. 28 12 4
2. 14 14 10
3. 46 44 44
4. 54 46 28
5. 52 50 40
6. 48 47 40
```

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TEST DATA

TEST INPUT:

1. A R R B B B G G O
   G R A B B R A G O O
   A R B G O B G A
2. AO AO BG GR BG OB OB OB AO
   BG BG BG BG AO AO AO GR OB OB
   BG GR AO AO GR BG OB OB
3. B G O R A B G O O R A B G O R A B G O R A
   AO BG GR OB AO BG GR B A O BG GR OB
   B R A G B OB R A G B O G AO BG BG AO GR
4. AO GR GR OB O B OB BG B G BG BG AO G R GR OB BG BG
   AO OB G R BG AO O B GR BG AO OB GR BG A GR BG AO OB GR BG
   AO OB GR BG AO OB G R BG R A G OB B AO
5. AO OB GR BG A B R G O
   OB GR BG AO B R G O A
   GR BG AO OB R G O A B
6. O BG GR OB OB B A GR BG BG BG OB
   O G A GR OB AO BG BG B G AO OB OB OB BG
   B B B B B B B B B B B B B B B

TEST OUTPUT:

1. 30 28 24
2. 80 70 60
3. 85 79 59
4. 134 127 85
5. 44 44 44
6. 101 90 90

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测试数据

测试输入：

1. A R R B B B G G O
   G R A B B R A G O O
   A R B G O B G A
2. AO AO BG GR BG OB OB OB AO
   BG BG BG BG AO AO AO GR OB OB
   BG GR AO AO GR BG OB OB
3. B G O R A B G O O R A B G O R A B G O R A
   AO BG GR OB AO BG GR B A O BG GR OB
   B R A G B OB R A G B O G AO BG BG AO GR
4. AO GR GR OB O B OB BG B G BG BG AO G R GR OB BG BG
   AO OB G R BG AO O B GR BG AO OB GR BG A GR BG AO OB GR BG
   AO OB GR BG AO OB G R BG R A G OB B AO
5. AO OB GR BG A B R G O
   OB GR BG AO B R G O A
   GR BG AO OB R G O A B
6. O BG GR OB OB B A GR BG BG BG OB
   O G A GR OB AO BG BG B G AO OB OB OB BG
   B B B B B B B B B B B B B B B

测试输出：

1. 30 28 24
2. 80 70 60
3. 85 79 59
4. 134 127 85
5. 44 44 44
6. 101 90 90